

runs to make sure the conclusion is reliable.



(a) Caption: *The hair drier is facing away from the person.* Label: False.



(b) Caption: *The bench is in front of the person.* Label: True.

Figure 9: LXMERT failed on both examples.

Performance by relation. We give performance by relation for all three finetuned models on both random and zero-shot splits in Fig. 8. The order from left to right is sorted by the frequency of relations in the dataset (within each split). Interestingly, there does not seem to be any correlation between performance and frequency of the relation, hinting that specific relations are hard not due to an insufficient number of training examples but because they are fundamentally challenging for current VLMs. Any relation that requires recognising orientations of objects seems to be hard, e.g., “facing”, “facing away from”, “parallel to” and “at the back of”. As an example, LXMERT failed on the two examples in Fig. 9 which require understanding the front of a hair drier and a person respectively. In this regard, left-right relations such as “at the left/right side of” and “left/right of” are difficult because the intrinsic reference frame requires understanding the orientation of objects. As an example, in Fig. 1, all three models predicted False, but in the intrinsic frame (i.e., from the bench’s point of view), the potted plant is indeed at the right.

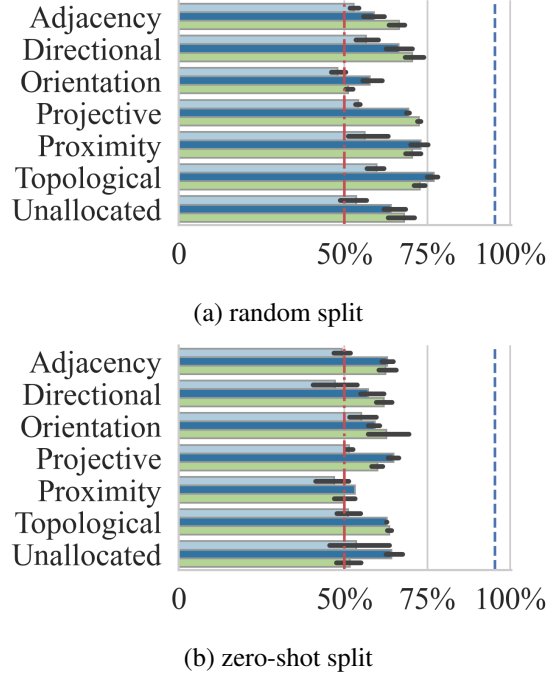


Figure 10: Performance by categories of relations, on the random and zero-shot split test sets. For legend information, see Fig. 8.

To get a more high-level understanding of the relations’ performance, we group model performance by the categories of Marchi Fagundes et al. (2021): “Adjacency”, “Directional”, “Orientation”, “Projective”, “Proximity”, “Topological” and “Unallocated” (also shown in Table 1). The results are shown in Fig. 10. “Orientation” is the worst performing group on the random split, and on average all models’ performances are close to the chance level. When comparing random and zero-shot splits, performance has declined to some extent for almost all categories and models. The decrease in “Proximity” is particularly drastic across all models – it declined from close to 75% accuracy in random split to chance level in zero-shot split. “Proximity” contains relations such as “close to”, “near” and “far from”. We believe it is due to the fact that the notion of proximity is relative and very much dependent on the nature of the concept and its frequent physical context. E.g., for a “person” to be “near” an indoor concept such as “oven” is very different from a person being “near” a frequent outdoor object such as “train” or “truck”. Since the zero-shot split prevents models from seeing test concepts during training, the models have a poor grasp of what counts as “close to” or “far from” for these concepts, thus generalising poorly.



Figure 11: Caption: *The cow is at the back of the car.* Label: True. LXMERT and VisualBERT predicted False.

Other Errors. While certain relations are intrinsically hard, we have observed other types of errors that are not bounded to specific relations. Here we give a few examples. Some instances require complex reasoning. In Fig. 11, the model needs to recognise that both the cow and the back of the car are in the car’s side mirror and also infer the relative position of the back of the car and the cow. It is perhaps no surprise that two of the three models predicted wrongly. Some other examples require common sense. E.g., in Fig. 2, we can infer the person and the cow’s moving direction and can then judge if the cow is ahead of the person. LXMERT failed on this example. In Fig. 3 (right), the model needs to infer that the main body of the cat is hidden behind the laptop. Interestingly, all three models predicted this example correctly.

5.3 Case Study on Reference Frames

As discussed in §4.3, different frames of reference can be used in natural language and it would be helpful to understand whether our models recognise them. We argue that the task of identifying frame of reference itself is very hard for current models. However, learning to recognise frames of reference helps the task of visual spatial reasoning.

Firstly, we conduct a case study on left/right-related relations. We additionally label the reference frames of all true statements containing any of the left/right-related relations. We exclude all data points that can be interpreted in both intrinsic and relative frames to slightly reduce the complexity of the task. Then we finetune a ViLT checkpoint to predict the reference frame based on the true statement and the image. The model’s performance on test set is shown in the upper half of Table 5. We can see that reference frame prediction is an extremely hard task for the model. This is presum-

<i>Reference frame prediction task</i>			
model↓	Precision	Recall	F1
ViLT	59.2 $\pm$ 3.7	59.7 $\pm$ 5.8	56.9 $\pm$ 4.4
<i>VSR task (left/right subset)</i>			
model↓	Accuracy		
ViLT	54.2 $\pm$ 0.6		
ViLT + rf_trained	59.2 $\pm$ 1.8		

Table 5: ViLT model performance on the reference frame prediction task (upper half; we report macro-averaged Precision/Recall/F1 since the binary classification task is imbalanced); and VSR task using original pretrained checkpoint or the reference frame prediction task trained checkpoint (accuracy reported).

ably because it requires taking into account a 3D viewpoint and simulating transformations between different viewpoints.

Secondly, we use this model trained with reference frame labels to initialise the VSR task model and further finetune it on the VSR task (only the left/right relations). The test results are shown in the lower part of Table 5.⁸ We see a clear positive transfer from reference frame prediction task to the VSR task. This suggests that learning to recognise reference frames can indeed help downstream visual spatial reasoning. This makes sense since simulating the transformation of intrinsic/relative frames could be an intermediate reasoning step in detecting whether a statement is true/false.

5.4 Sample Efficiency

In order to understand the correlation between model performance and the number of training examples, we conduct sample efficiency analysis on VSR. The results are plotted in Fig. 12. For the minimum resource scenario, we randomly sample 100 shots from the training sets of each split. Then we gradually increase the number of training examples to be 25%, 50% and 75% of the whole training sets. Both LXMERT and ViLT have a reasonably good few-shot capability and can be quite performant with 25% of training data. LXMERT, in particular, reaches above 55% accuracy with 100 shots on both splits. The zero-shot split is substantially harder

⁸Note that the reference frame train/dev/test sets are derived from the VSR task split – so no data leakage is possible from train to dev and test sets even after the intermediate pretraining.

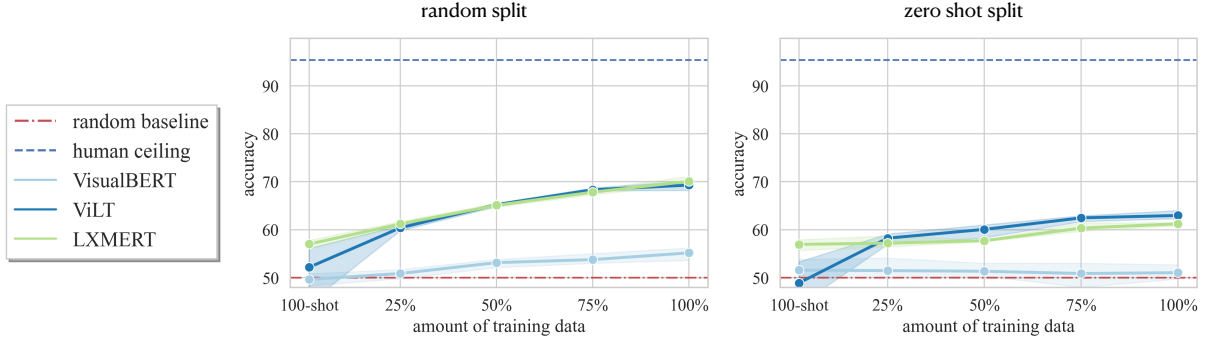


Figure 12: Sample efficiency analysis: model performance under different amounts of training data (100-shot, 25%, 50%, 75% and 100% of training set). Results on both the random and zero-shot split test sets are shown. As training data increases, the performance plateaus on both sets but the flattening trend is more obvious on the zero-shot split.

and most models appear to have already plateaued at around 75% of the training set. For the random split, all models are increasing performance with more data points, though improvement slows down substantially for LXMERT and ViLT after 75% of training data. The fact that LXMERT has the best overall few-shot capability may be suggesting that LXMERT’s pretrained object detector has a strong inductive bias for the VSR dataset as it does not need to learn to recognise concept boundaries and classes from scratch. However, this advantage from LXMERT seems to fade away as the number of training examples increases.

6 Conclusion and Future Directions

We have presented Visual Spatial Reasoning (VSR), a controlled probing dataset for testing vision-language models (VLMs)’ capabilities of recognising and reasoning about spatial relations in natural image-text pairs. We made a series of linguistic observations on the variability of spatial language when collecting VSR. We highlighted the diverse use of reference frames among annotators, and also the ambiguous nature of certain spatial relations. We tested four popular VLMs on VSR, and found they perform more than 25% below the human ceiling. On a more challenging concept zero-shot split, the tested VLMs struggled to reach 60% accuracy and their performance plateaued even with increased training examples. Among the finetuning-based VLMs, ViLT and LXMERT outperformed VisualBERT, and we pointed out that the explicit positional information in the former two models is crucial in the task. CLIP with prompt engineering achieved slightly better than random performance,

suggesting poor capability in spatial reasoning. We also performed a by-relation analysis and found that the models’ performances on certain relations have little correlation with the number of training examples, and certain relations are inherently more challenging. We identified orientation as the most difficult category of relations for VLMs. Proximity is another challenging category, especially in the zero-shot setup as this relation is highly concept-dependent. We hope the task serves as a useful tool for testing and probing future VLMs.

In future work, we plan to more extensively investigate whether large-scale pretrained dual-encoders such as CLIP (Radford et al., 2021), ALIGN (Jia et al., 2021) and LiT (Zhai et al., 2022) can properly recognise spatial relations, especially in the finetuning setup. A comparison of dual- and cross-encoders’ performance on each spatial relation might guide future model design. Recently, Alayrac et al. (2022); Chen et al. (2023); Huang et al. (2023) proposed ultra-large-scale VLMs. It would be interesting to see if VLMs have better spatial reasoning capability when scaled up. Another direction is extending VSR to cover more languages and cultures (Liu et al., 2021; Bugliarello et al., 2022) and test multilingual VLMs. Along the same line, since we have also collected the meta-data of annotators, the VSR corpus can be used as a resource for investigating research questions such as: How is “space” described among different dialects of English? How is “space” perceived among different populations? We hope that the annotation process of VSR can also serve as a basis for future cross-lingual and cross-cultural sociolinguistic research.